

Note on: *Bond Total Return Swaps — Theory, Pricing & Practice*

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Reading notes

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One-paragraph summary

A practitioner write-up of bond TRS pricing in three components: bond coupons, bond performance (change in price), and the funding leg, all discounted with a *risky discount factor* $\tilde{P}(t_0, t_i) = Q(t_0, t_i) P(t_0, t_i)$ that combines a survival probability Q and a risk-free DF P , plus a loss-given-default (LGD) term to account for unwind on bond default. The PV identity decomposes into four summations (coupons + performance – funding – LGD), each carrying the same risky DF. Future bond prices are projected at the repo rate, $B_T = B_t(1 + r \tau_1 - r c \tau_2)$, with repo interest deducted on intermediate coupons. The *par spread* p is defined as the funding spread that makes PV zero; rearranging gives $p = \text{PV}(\text{TRS}|s = 0)/\text{RiskyAnnuity}^{\text{float}}$. The appendix offers a compact alternative valuation: net the TRS as “full bond cashflows to bond maturity + funding leg to TRS maturity + a negative notional exchange equal to the current bond price paid at TRS maturity”, useful for investment-grade bonds where the LGD term is small.

1 Setup and notation

Paper	House	Meaning
ϕ	$\phi \in \{+1, -1\}$	+1 receiver of bond coupons, -1 payer
N_B	—	Bond notional = Units $\times$ Face Value
N_C	—	Cash notional of the funding leg
r	—	Bond coupon rate (fixed)
τ_i	α_i	Year fraction of bond coupon period i
τ_j	α_j	Year fraction of funding-leg period j
F_j	—	Floating funding index fixing (e.g. SOFR) at j
s	—	Funding spread (added to F)
p	—	Par / breakeven funding spread
$B(t)$	$P_t(\cdot)$	Dirty bond price at t
$Q(t_0, t_i)$	$Q(t_0, t_i)$	Survival probability from t_0 to t_i
$P(t_0, t_i)$	D_{t_0, t_i}	Risk-free discount factor
$\tilde{P}(t_0, t_i)$	—	Risky DF = $Q \cdot P$
RR	—	Recovery rate (paper convention: 40% senior subordinated)
$1 - RR$	—	LGD per unit notional
r^{repo}	r^{repo}	Repo rate (bond-specific)
c	—	Coupon paid during projection window
τ_1, τ_2	—	Time from t to T ; time from coupon date to T

2 Key equations

TRS PV (top-level structure):

$$\text{PV}_{\text{TRS}} = \phi \left(\text{PV}(\text{Bond Coupons}) + \text{PV}(\text{Performance}) - \text{PV}(\text{Funding}) \right), \quad (1)$$

with $\phi = +1$ if receiving bond coupons (paying funding), -1 otherwise. LGD added separately as shown below.

PV of bond coupons.

$$\text{PV}(\text{Bond Coupons}) = \sum_{i=1}^n N_B r \tau_i \underbrace{Q(t_0, t_i) P(t_0, t_i)}_{=\tilde{P}(t_0, t_i)}. \quad (2)$$

Says: each fixed coupon $N_B r \tau_i$ is discounted by the risky DF combining survival to t_i and risk-free discounting from t_0 to t_i .

PV of bond performance — two settlement conventions. Performance paid on each coupon date:

$$\text{PV}(\text{Performance}) = \sum_{i=1}^n N_B (B(t_{i-1}) - B(t_i)) Q(t_0, t_i) P(t_0, t_i). \quad (3)$$

Performance paid at maturity T only:

$$\text{PV}(\text{Performance}) = N_B (B(t_0) - B(T)) Q(t_0, T) P(t_0, T). \quad (4)$$

Says: performance is the running change in bond price. The periodic version reduces PFE and XVA costs (Trade Feature #9) at the expense of more frequent operational settlement.

PV of the funding leg.

$$\text{PV}(\text{Funding}) = \sum_{j=1}^m N_C (F_j + s) \tau_j Q(t_0, t_j) P(t_0, t_j). \quad (5)$$

Says: floating index F plus spread s , accruing on cash notional N_C . The same survival weighting applies because the TRS unwinds on bond default — the funding leg also stops.

Full TRS PV with LGD.

$$\begin{aligned} \text{PV}_{\text{TRS}} = \phi \left[\sum_{i=1}^n N_B r \tau_i \tilde{P}(t_0, t_i) + \sum_{i=1}^n N_B (B(t_{i-1}) - B(t_i)) \tilde{P}(t_0, t_i) \right. \\ \left. - \sum_{j=1}^m N_C (F_j + s) \tau_j \tilde{P}(t_0, t_j) - \sum_{i=1}^n N_B (1 - RR) (Q(t_0, t_{i-1}) - Q(t_0, t_i)) P(t_0, t_i) \right]. \end{aligned} \quad (6)$$

Says: the fourth summation is the LGD term. Default in $[t_{i-1}, t_i]$ has probability $Q(t_0, t_{i-1}) - Q(t_0, t_i)$; multiplied by the (risk-free) discount and the LGD per unit notional.

Bond forward via repo (Burgess uses simple compounding):

$$B_T = B_t (1 + r^{\text{repo}} \tau). \quad (7)$$

With intermediate coupons c deducted at the repo rate from coupon date to T :

$$B_T = B_t (1 + r^{\text{repo}} \tau_1) - c (1 + r^{\text{repo}} \tau_2) \Big|_{\text{summed over coupons in } [t, T]}. \quad (8)$$

Says: the dirty bond is treated as “on deposit” earning the repo rate. Intermediate coupons accrete to T at the same rate. This is the simple-compounding analogue of Pucci’s $\text{Fwd}_t(t_e) = P_t^{\text{mkt}} e^{r^{\text{repo}}(t_e - t)} - \sum c_i e^{r^{\text{repo}}(t_e - t_i)}$.

Par (breakeven) funding spread.

$$p = \frac{\text{PV}(\text{TRS} | s = 0)}{\sum_j N_C \tau_j \tilde{P}(t_0, t_j)}, \quad \text{Risky Annuity (float leg)} = \sum_j N_C \tau_j \tilde{P}(t_0, t_j). \quad (9)$$

Says: solve $\text{PV} = 0$ for s ; p is the spread that makes the two legs of equal value. The denominator is the risky annuity of the funding leg.

Approximate decomposition (Appendix):

- Full bond cashflows to bond maturity.
- Funding leg to TRS maturity.
- A negative notional exchange equal to the current bond price paid at TRS maturity — this is what represents the short performance leg in compact form.
- For investment-grade bonds, set $Q \equiv 1$ and drop LGD.

3 Derivations & commentary

Commentary

The note is operational, not theoretical. The PV structure (6) treats every cashflow under the *same* risky DF $\tilde{P}$, which is internally consistent given the “TRS unwinds on default” clause: if the bond defaults, both legs stop simultaneously and an LGD payment

closes out the position. The single-curve setup hides several subtleties that the post-2008 Lou (2018) paper makes explicit (separate repo for the bond hedge vs. collateral/OIS for derivative discounting).

Commentary

Two notational conflicts to watch when reading alongside other notes: **(i)** Burgess uses ϕ for the long/short flag where Pucci uses a and others use θ . **(ii)** Q is the survival probability here, not (as in some credit-derivatives texts) the risk-neutral measure. The risky DF $\tilde{P} = Q \cdot P$ is then the credit-flavoured analogue of Pucci's pre-default $\widehat{D}_{t,T}$ under default-independence (DI) and absolute-continuity (AC); they coincide under DI with $Q(t, T) = e^{-\int_t^T \gamma_u du}$.

Commentary

The repo-projection formula (7)–(8) is the simple-compounding version of what the anonymous Treasury Lock Model and Pucci 2019 derive in continuous form. For short tenors ($\tau \lesssim 1Y$) the simple form is fine; for longer projections the compound version (or the continuous $e^{r\tau}$ used in the T-Lock paper) drifts apart and the choice matters.

Commentary

“Constant Units” vs. “Constant Notional” is a contract feature, not a pricing distinction. Under *Constant Units* the bond quantity is fixed and the cash notional varies (periodic notional exchanges required, bond fixings needed). Under *Constant Notional* the cash side is fixed and the bond quantity drifts. Pricing uses the same building blocks; only the bookkeeping of N_B vs. N_C in successive periods changes.

Commentary

The appendix's compact valuation (“bond + funding leg + negative notional payment at maturity”) is useful because it lets one re-use existing bond-pricer and IRS-pricer without building dedicated TRS code. For investment-grade names where LGD is small it's accurate enough to be the production fallback. The key insight is that the receiver of performance is essentially long the bond *minus* the bond price paid at TRS maturity, which is exactly the payoff of buying the bond now and selling it at T at the forward price.

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

Quantities the paper defines:

- The risky discount factor $\tilde{P}(t_0, t_i) = Q(t_0, t_i) P(t_0, t_i)$.
- Four PV components: bond coupons (2), bond performance (periodic (3) or bullet (4)), funding leg (5), and the LGD term in (6).
- Forward bond price under simple repo compounding (7)–(8).
- The par funding spread p via (9) and the risky annuity of the float leg.

Canonical numerical case (pages 7–9):

- 50,000 bonds purchased at 104.5433%; cash equivalent USD 5,227,165.
- Bond leg specified in Units.
- Funding: SOFR + 100 bps.
- Performance paid at maturity only.
- Recovery rate $RR = 40\%$ (paper assumption for senior subordinated).
- Coupon worked example: $80,736 \times 100 \times 3.85\% \times 0.5 = 155,416.80$; Units = $10,000,000 / (100 \times 123.86\%) = 80,736$.

Expected numerical values: no PV bottom-line printed in the document — only the per-period coupon arithmetic above. The formulas are self-validating by structure.

Equation $\leftrightarrow$ quantity map:

Paper eq.	Symbol	Quantity
(2)	PV(Coupons)	risky-discounted bond coupon stream
(3)	PV(Performance)	periodic bond-price change PV
(4)	PV(Performance)	bullet (maturity-only) performance PV
(5)	PV(Funding)	risky-discounted floating coupon PV
(6)	PV_{TRS}	full PV with LGD
(7)–(8)	B_T	forward bond price via simple repo
(9)	p	breakeven funding spread

Invariants and identities to verify:

1. *Risky DF factorisation:* $\tilde{P}(t_0, t_i) = Q(t_0, t_i) P(t_0, t_i)$ holds by definition; in particular $\tilde{P} \rightarrow P$ as $Q \rightarrow 1$ (no default).
2. *Par spread reproduction:* pricing the TRS at spread $s = p$ from (9) returns $PV_{\text{TRS}} = 0$ within float tolerance.
3. *Periodic vs. bullet performance equivalence:* in the zero-default-zero-discounting limit ($Q \equiv 1, P \equiv 1$), $\sum_i (B(t_{i-1}) - B(t_i)) = B(t_0) - B(T)$ telescopes exactly; the two performance formulations differ only by survival/discount weights on intermediate periods.
4. *Investment-grade limit:* with $Q \equiv 1$, the LGD term vanishes (because the survival differences are zero) and (6) reduces to a plain bond-vs-floater swap.
5. *Coupon arithmetic* (worked example): $80,736 \times 100 \times 3.85\% \times 0.5 = 155,416.80$, and $10,000,000 / (100 \times 123.86\%) = 80,736$ (rounded).
6. *Default-period sum:* $\sum_{i=1}^n (Q(t_0, t_{i-1}) - Q(t_0, t_i)) = Q(t_0, t_0) - Q(t_0, t_n) = 1 - Q(t_0, t_n)$ (telescoping); used in the LGD term.
7. *Repo forward consistency:* (7) is the first-order Taylor approximation of $B_T = B_t e^{r\tau}$ (Pucci's continuous form, eq. 50); the relative error at $r = 5\%$, $\tau = 1Y$ is $\sim 0.12\%$. For TRS tenors beyond a few years, the compound form should be used.

5 Glossary of symbols (paper's notation)

ϕ	+1 receiver of bond / -1 payer of bond
N_B	Bond notional, Units $\times$ Face Value
N_C	Cash notional of the funding leg
r	Fixed bond coupon rate
τ_i	Year fraction of coupon period i
τ_j	Year fraction of funding period j
F_j	Floating funding index fixing at j (e.g. SOFR)
s	Funding spread (added to F)
p	Par / breakeven funding spread
$B(t)$	Dirty bond price at t
$Q(t_0, t_i)$	Survival probability from t_0 to t_i
$P(t_0, t_i)$	Risk-free discount factor from t_0 to t_i
$\tilde{P}(t_0, t_i)$	Risky discount factor = $Q \cdot P$
RR	Recovery rate (paper assumes 40% senior subordinated)
$1 - RR$	LGD per unit notional
r^{repo}	Repo rate of the underlying bond
c	Coupon amount paid during a projection window
τ_1, τ_2	Times: $t \rightarrow T$ and coupon date $\rightarrow T$
T	TRS maturity date

TODO / Open questions

- Build a worked end-to-end PV computation on the case-study inputs (50,000 units at 104.5433%, SOFR + 100 bps, $RR = 40\%$, sensible curve). The paper sets up the inputs but does not print a bottom-line PV.
- Cross-check the simple-repo forward (7) against the continuous form $B_t e^{r\tau}$ for $\tau \in \{0.5, 1, 2, 5\}Y$ at $r^{\text{repo}} \in \{2\%, 5\%\}$ — quantify the gap.
- Implement the approximate decomposition in the appendix and compare with the full (6) for an investment-grade name (low $1 - Q$); the gap should be the LGD term.
- Par-spread sensitivity to recovery rate at fixed curve and survival: should depend on RR only through the LGD term.
- Reconcile the Burgess setup (single risky DF for both legs) with the Lou (2018) “security leg + premium leg under different discounting” picture. They should agree under Burgess’s assumption that the TRS unwinds at default; they will disagree if the funding leg discounting is taken to be the dealer’s funding rate.